

Lecture #9 / October 1, 2026

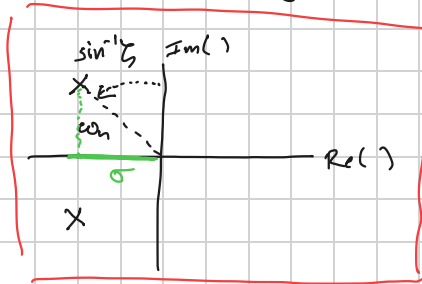
Problem Set #2 Due Next Thursday

Recap of Last Lecture

Time-domain Specs $\rightarrow$ set of requirements on the performance of a system

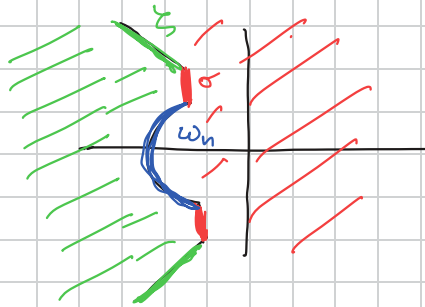
Four Different Specs:

- ① Rise Time
- ② Overshoot
- ③ Peak Time
- ④ Settling Time



$$\begin{aligned} t_r &\approx \frac{1.8}{\omega_n} \\ t_p &= \frac{\pi}{\omega_d} \rightarrow \omega_d = \omega_n \sqrt{1-\zeta^2} \\ M_p &= \exp \left\{ \frac{-\pi \zeta}{\sqrt{1-\zeta^2}} \right\} \\ t_s &\approx \frac{4.6}{\sigma} \leftarrow \text{Real part of the pole} \end{aligned}$$

Can derive constraint regions for the poles



Geist: I give you performance specs, like rise, overshoot, etc.

You tell me where the poles need to go to satisfy the requirements

General Rules of Thumb:

- ① A "nearby" real LHP zero increases overshoot
- ② A real RHP zero cause an inverse response (decreases overshoot, but it also creates undershoot)
- ③ An extra LHP pole slows the rise time and as it approaches the origin, the dominant mode "looks" more like a first-order system

A Different Concept: Stability

Previously, we wanted to know where do we place our poles to guarantee performance spec's.

Now, we want to know

- ① Where do we place our poles to ensure a bounded / convergent response?
LHP
- ② How do we place our poles such that the response is bounded / convergent?
Routh Criterion
(How $\Rightarrow$ How affect the characteristic polynomial through a choice of controller gains?)

IMPORTANT:

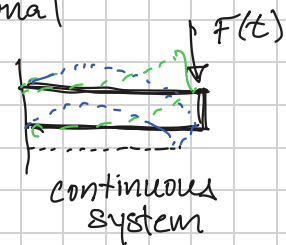
A finite-dimensional FTI system is asymptotically stable if and only if all roots of its characteristic polynomial have strictly negative real parts

Finite-Dimensional vs Infinite Dimensional

$\rightarrow$ ODE's

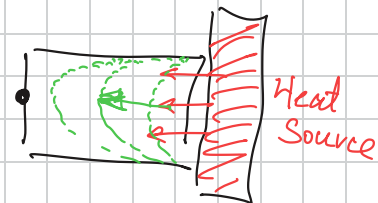
$\rightarrow$ PDE's

$\rightarrow$ Beam

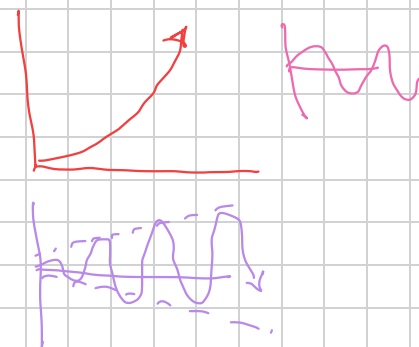
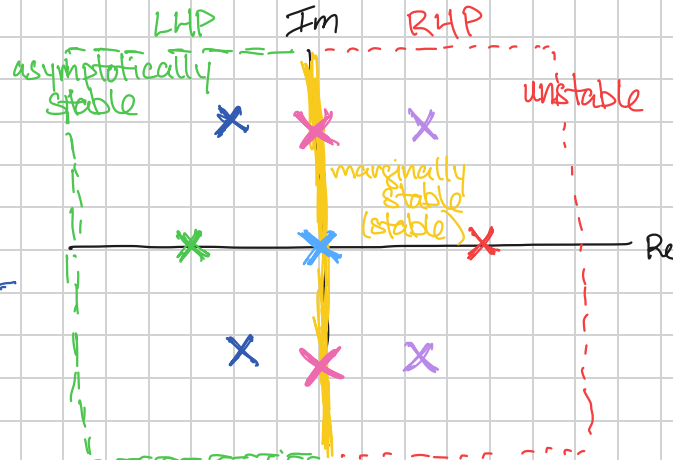
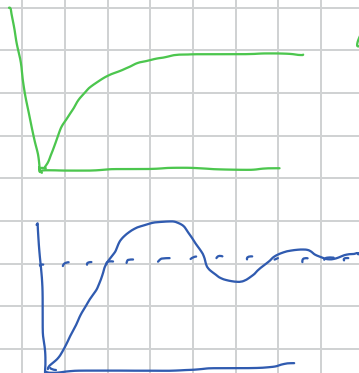
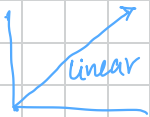


You can "sample" a PDE to convert it into a system of ODE's

$\rightarrow$ Temperature of a body



$\frac{1}{s}$



BIBO Stability (Bounded Input / Bounded Output)

Q: Does every bounded input produce a bounded output?

A: Start w/ a causal, strictly proper system with impulse response $h(t)$

Let's say that input $|u| \leq M$

$$\Rightarrow |y(t)| = \left| \int h(\tau) u(t-\tau) d\tau \right| \leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

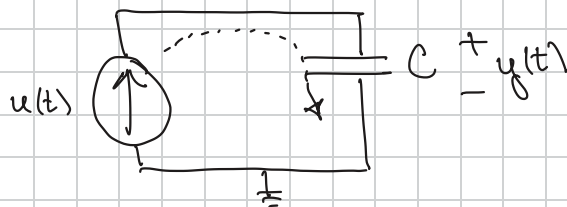
Basic Convolution

$\Rightarrow$ The output is bounded whenever $\int |h(\tau)| d\tau$ is bounded.

$\rightarrow$ Definition of BIBO stability

$$\text{BIBO stable} \iff \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Example:



$$C \dot{v} = i$$

$$\Rightarrow C s \bar{V}(s) = \bar{I}(s)$$

$$\Rightarrow \frac{\bar{V}(s)}{\bar{I}(s)} = \frac{1}{sC}$$

Assume $C = 1$, then

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau = \int_{-\infty}^{\infty} 1 d\tau = \infty$$

$\leftarrow$ $\underline{= 1(t)}$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = h(t) = 1(t)$$

$\Rightarrow$ This system is NOT BIBO stable.

Internal Stability

$$a(s) = s^n + a_1 s^{n-1} + \dots + a_n = 0$$

distinct
roots
ONLY

$$\Rightarrow y(t) = \sum_{i=1}^n k_i e^{p_i t} \quad \text{where } k_i \text{ is scaling factor}$$

p_i is pole of char. poly.

Decay for every set of initial cond.

if and only if $\operatorname{Re}\{p_i\} < 0$ for all i

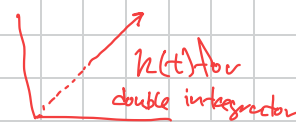
This is internal asymptotic stability.

~~Distinct Roots~~

$\Rightarrow$ We can lose internal A.S.

$\rightarrow$ Repeated poles $\Rightarrow \underline{te^{*j\omega t}}$
unbounded

e.g., double integrator $\rightarrow$



"Cancellation Trap"

"If a zero were to cancel a pole in the RHP, the corresponding k_i would equal zero in the output, but the unstable transient would appear in an "internal variable"

e.g., $G(s) = \frac{1}{s-1}$ (PLANT) $C(s) = \frac{s-1}{s+3}$ (CONTROLLER)

$$T = \frac{GC}{1+GC}$$

$$G(s)C(s) = \frac{1}{s+3} \Rightarrow T = \frac{1/s+3}{1+1/s+3} = \frac{1}{s+4}$$

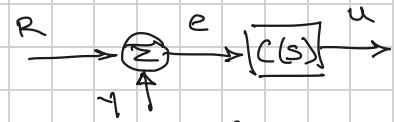
stable transfer function

BUT: it doesn't guarantee internal asymptotic stability!

① Split the controller: $C(s) = \frac{s-1}{s+3} = 1 + \frac{-4}{s+3}$

$$\dot{x}_c = -3x_c + e$$

$$u = e - 4x_c$$



$$U(s) = C(s)E(s)$$

$$U(s) = \left(1 + \frac{-4}{s+3}\right)E(s)$$

$$u(s) = E(s) + \frac{-4}{s+3}E(s)$$

Step 2: $e = r - y = r - x_p$

Plant is $\frac{1}{s-1}$

$$\Rightarrow \dot{x}_p = x_p + u$$

$$(s+3)U(s) = E(s)(s+3) - 4E(s)$$

$$\dot{u} + 3u = \dot{e} + 3e - 4e$$

$$\Rightarrow \dot{u} + 3u = \dot{e} - e$$

$$\dot{x}_p = x_p + (r - x_p) - 4x_c = r - 4x_c$$

$$\dot{x}_c = -3x_c + r - x_p$$

$$\dot{x} = Ax + Br \quad \text{where } A = \begin{bmatrix} 0 & -4 \\ -1 & -3 \end{bmatrix}$$

$$\det(sI - A) = (s+4)(s-1)$$

$\Rightarrow$ NOT internally stable

ARE THE CLOSED LOOP MODES

Routh's Criterion is an algorithm for determining how many poles of a characteristic polynomial have POSITIVE real parts.

We use this for determining whether our closed-loop system is asymptotically stable (or not)

① A necessary condition for stability is that every coefficient of $a(s)$ [char poly] is present and positive non-zero

② The Routh Array:

s^n	1	a_2	a_4	...
s^{n-1}	a_1	a_3	a_5	...
s^{n-2}	b_1	b_2	b_3	
$\vdots$	c_1	c_2	c_3	
1				

$$a(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

$$b_1 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_2 \\ a_1 & a_3 \end{bmatrix}$$

$$b_2 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_4 \\ a_1 & a_5 \end{bmatrix}$$

$$b_3 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_6 \\ a_1 & a_7 \end{bmatrix}$$

$\vdots$

$$c_1 = -\frac{1}{b_1} \det \begin{bmatrix} a_1 & a_3 \\ b_1 & b_2 \end{bmatrix}$$

$$c_2 = -\frac{1}{b_1} \det \begin{bmatrix} a_1 & a_5 \\ b_1 & b_3 \end{bmatrix}$$

$\vdots$

$$c_i = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

IMPORTANT:

All roots are strictly in the LHP i.e. if the array has positive first column.

AND The number of RHP roots equals the number of sign changes in the left column

e.g. 1

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

s^6	1	3	1	4
s^5	4	2	4	0
s^4	2.5	0	4	
s^3	2	-2.4	0	
s^2	3	4		
s^1	-5.1	0		
1	4			

Everything needs to be positive

⇒ this char poly has poles in RHP

⇒ 2 poles in the RHP b/c of sign changes